25/9/26

Activity 2.3 $\frac{dy}{dt} = f(t,y) = F(t)g(y)$?

$$\frac{dy}{dt} = \frac{ty^2 - t^2}{y^2} \qquad y \neq 0$$

$$\therefore f(t,y) = \frac{ty^2 - t^2}{y^2} = F(t)g(y) \text{ ?}$$

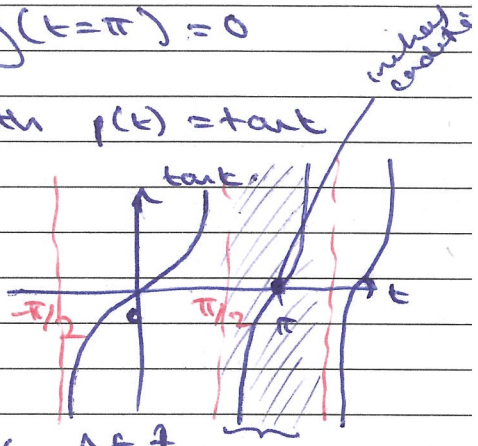
a b c d

$f(1,1)f(4,2) = 0$
$f(1,2)f(4,1) = -9$ } ∴ not separable.

Example 2.11

$$\frac{dy}{dt} + (\tan t)y = \sin t \qquad y(t=\pi) = 0$$

This is first order ODE that is linear with $p(t) = \tan t$ and $g(t) = \sin t$



$g(t) = \sin t$ is continuous $\forall\, t \in \mathbb{R}$

$p(t) = \tan t = \frac{\sin t}{\cos t} \longrightarrow \infty$

" 0 @ $t = \pi/2, 3\pi/2, \ldots$

$p(t)$ has discontinuities @ $t = (2n+1)\pi/2 \quad n \in \mathbb{Z}$

The initial condition $y(t_0 = \pi) = 0$ has $\pi/2 < t_0 < 3\pi/2$

By theorem 2.6 ∃ unique solution to the IVP for all $t \in (\pi/2, 3\pi/2)$ //

Plan

- first order ODEs – linear (IF), separable, homogeneous.
  – Existence & uniqueness.
- Examples sheet 1 – release 10th on Monday.
- Tutorial sheet 1 & feedback sheet 1 – next week / week 3

Help Sessions: Monday @ 1230 H206
LSI / Sem B @ 430.

Proof Theorem 3.3

we are required to show that

$$y_3'' + p y_3' + q y_3 = 0 \qquad (*)$$

we know that $y_3 = c_1 y_1 + c_2 y_2$ $\qquad y_1 = y_1(t),\ y_2 = y_2(t)$

$$\Rightarrow y_2' = c_1 y_1' + c_2 y_2'$$
$$y_3'' = c_1 y_1'' + c_2 y_2''$$

Now substitute into LHS of (*)

$$c_1 y_1'' + c_2 y_2'' + p(c_1 y_1' + c_2 y_2') + q(c_1 y_1 + c_2 y_2)$$

$$= c_1 \underbrace{(y_1'' + p y_1' + q y_1)}_{=0 \text{ since } y_1 \text{ satisfies } (*)} + c_2 \underbrace{(y_2'' + p y_2' + q y_2)}_{=0 \text{ by } (*)}$$

$$= 0$$

$$\therefore y_3'' + p y_3' + q y_3 = 0 \text{ as required } \blacksquare$$

Proof of Theorem 3.6

we are required to show that

$$W = y_1 y_2' - y_2 y_1' \qquad (\text{def}^n\ 3.4)$$
$$= c \exp\left(-\int p(t)\,dt\right)$$

we know that

$$y_1'' + p y_1' + q y_1 = 0 \qquad (1)$$
$$y_2'' + p y_2' + q y_2 = 0 \qquad (2)$$

Now compute $y_1 \times (2) - y_2 \times (1)$:

$$y_1 y_2'' + p y_1 y_2' + q y_1 y_2 - y_2 y_1'' - y_2 p y_1' - y_2 q y_1 = 0$$

$$y_1 y_2'' - y_2 y_1'' + p\underbrace{(y_1 y_2' - y_2 y_1')}_{W} = 0$$

$$\Rightarrow (y_1 y_2')' - y_2' y_1 - (y_2 y_1')' + y_1' y_2 + pW = 0$$

$$\Rightarrow (y_1 y_2' - y_2 y_1')' + pW = 0$$

$$\Rightarrow W' + pW = 0$$

This is a separable eqⁿ for W.

$$\frac{dW}{dt} = -pW$$

$W=0$ is a special soln

For $W \neq 0$ we have the typical solution from

$$\int \frac{dW}{W} = -\int p\,dt \qquad \text{where } p = p(t)$$

$$\Rightarrow \ln|W| = -\int p\,dt + c_1$$

$$\Rightarrow e^{\ln|W|} = e^{-\int p\,dt + c_1} = e^{-\int p\,dt} \cdot e^{c_1}$$

$$\Rightarrow |W| = e^{c_1} e^{-\int p\,dt}$$

$$\Rightarrow W = c\,e^{-\int p\,dt} \qquad \text{where } c = \pm e^{c_1}$$

as required.

Example 3.8

- $y_1 = \cos 3t \Rightarrow y_1' = -3\sin 3t \Rightarrow y_1'' = -9\cos 3t$

and to substituting into LHS of ODE gives

$$-9\cos 3t + 9\cos 3t = 0 \quad \checkmark$$

- Repeat for y_2

$\therefore y_1$ and y_2 are solutions.

- y_1 and y_2 are linearly independent solution if $W \neq 0\ \forall t$, since $p=0$ and $q=9$ are continuous $\forall t$, by Th^m 3.5.

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 3t & \sin 3t \\ -3\sin 3t & 3\cos 3t \end{vmatrix}$$

$$= 3\cos^2 3t + 3\sin^2 3t$$

$$= 3 \neq 0 \quad \therefore y_1 \text{ and } y_2 \text{ are L.I.}$$

and they form a fundamental set (Th^m 3.7)

- For Abel's identity we require

$$W = 3 = c\exp\left(-\int p\,dt\right) \qquad (\dagger)$$

Here $\int p\,dt = \int 0\,dt = c_1$ constant

$\therefore (\dagger)$ is satisfied if $3 = c\exp(cc_1)$ $\therefore$ take $c = 3e^{c_1}$